

INDEPENDENT PEER-REVIEW PACKET

# EC-012

## Closed Strip-Threshold Majorant

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*Cold Review Packet 7 of 20*

**Dylan M. Roy Mathematical Research Program**

*Prepared for independent validation - 27 July 2026*

**This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.**

## READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

## Part I - Exact Review Object

|                                |                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Packet</b>                  | 7 of 20                                                                                                                                                                                                   |
| <b>Object ID</b>               | EC-012                                                                                                                                                                                                    |
| <b>Object class</b>            | Exact scalar inequality                                                                                                                                                                                   |
| <b>Exact statement</b>         | For $C(\mu)=\beta+\mu+A/\mu+B/\mu^2$ and $\mu_{\text{hat}}=\sqrt{A}+(2B)^{1/3}$ , the declared closed majorant, factor-two bound after subtracting $\beta$ , endpoint exactness, and $A=B=0$ bypass hold. |
| <b>Internal terminal label</b> | CLOSED - EXACT CLOSED STRIP-THRESHOLD MAJORANT                                                                                                                                                            |
| <b>Terminal date</b>           | 2026-07-21                                                                                                                                                                                                |
| <b>Recommended expertise</b>   | Inequalities / optimization / dynamical-systems estimates                                                                                                                                                 |

### 1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

### 2. Evidence class and prior review

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Evidence</b>            | One-variable inequality, endpoint analysis, and degenerate-case bypass.                                           |
| <b>Prior review record</b> | Independent random and endpoint tests confirmed the displayed bound and its restricted factor-two interpretation. |
| <b>Binding exclusions</b>  | No dynamical strip theorem or Palm/probability consequence without separate hypotheses.                           |

### 3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No dynamical strip theorem or Palm/probability consequence without separate hypotheses.

#### 4. Object-specific hostile checks

- Derive the bound from  $\mu_{\text{hat}}$  without assuming  $A, B > 0$  when an endpoint vanishes.
- Verify factor two only after subtracting  $\beta$ .
- Check  $A=0$ ,  $B=0$ , and  $A=B=0$  separately.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

#### 5. Proof-obligation ledger

| Obligation         | Reviewer test                                                              | Initial state   |
|--------------------|----------------------------------------------------------------------------|-----------------|
| O1 Object identity | Definitions, ordering, normalization, and law match the statement.         | OPEN FOR REVIEW |
| O2 Hypotheses      | Every hypothesis used is stated and no stronger premise is hidden.         | OPEN FOR REVIEW |
| O3 Derivation      | Each algebraic, analytic, topological, or probabilistic step is valid.     | OPEN FOR REVIEW |
| O4 Quantifiers     | Uniformity, existence, endpoint, and asymptotic quantifiers are correct.   | OPEN FOR REVIEW |
| O5 Computation     | Any executable or interval result is reproducible and non-vacuous.         | OPEN FOR REVIEW |
| O6 Scope firewall  | The conclusion is not silently enlarged beyond the exact object.           | OPEN FOR REVIEW |
| O7 Contradictions  | No known correction, counterexample, or retained caveat defeats the claim. | OPEN FOR REVIEW |

## Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

### GP-CLS-PROP-015-v1.0 — PROPOSED TERMINAL CLOSURE OF THE CLOSED STRIP-THRESHOLD MAJORANT

Artifact ID: GP-CLS-PROP-015-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy approves or modifies the terminal statement

Canonical impact: NONE before approval

#### 1. Exact object card

##### OBJECT CLASS: DETERMINISTIC ONE-VARIABLE THRESHOLD MAJORIZATION

SOURCE CONTEXT: invariant-strip threshold for the declared compensated cubic–quartic truncated flow

PARAMETERS:  $A \geq 0$ ,  $B \geq 0$ ,  $\mu > 0$ , and an additive base term  $\beta$

OBJECTIVE:  $C(\mu) = \beta + \mu + A/\mu + B/\mu^2$

CLOSED CHOICE:  $\hat{\mu} = \sqrt{A + (2B)^{1/3}}$ , except the degenerate  $A=B=0$  case, which bypasses strip optimization

EVIDENCE CLASS: exact inequalities and one-variable minimization

##### PROPOSED TERMINAL LABEL: CLOSED — EXACT CLOSED STRIP-THRESHOLD MAJORANT

#### 2. Exact terminal statements

For  $A, B \geq 0$  and  $\mu > 0$ , define

$$C(\mu) = \beta + \mu + A/\mu + B/\mu^2.$$

For  $A+B > 0$  choose

$$\hat{\mu} = \sqrt{A + (2B)^{1/3}}.$$

Then

$$C(\hat{\mu}) \leq C_{\text{closed}},$$

where

$$C_{\text{closed}} = \beta + 2\sqrt{A + (3/2) \cdot 2^{1/3} B^{1/3}}.$$

Let  $C_{\text{min}} = \inf_{\{\mu > 0\}} C(\mu)$ . Then

$$C_{\text{closed}} - \beta \leq 2(C_{\text{min}} - \beta).$$

Thus the closed form is a valid sufficient threshold and is never more than a factor two above the optimum after subtracting the common base term.

Endpoint exactness:

- If  $B=0$  and  $A>0$ ,  $\hat{\mu}=\sqrt{A}$  is the exact optimizer and  $C_{\text{closed}}=C_{\text{min}}$ .
- If  $A=0$  and  $B>0$ ,  $\hat{\mu}=(2B)^{1/3}$  is the exact optimizer and  $C_{\text{closed}}=C_{\text{min}}$ .
- If  $A=B=0$ ,  $\inf_{\{\mu>0\}} C(\mu)=\beta$  as  $\mu \downarrow 0$ ; the strip criterion must bypass root-finding rather than substitute  $\hat{\mu}=0$  into a  $\mu>0$  formula.

### 3. Fresh second-line verification

Validity at  $\hat{\mu}$ :

$$A/\hat{\mu} \leq A/\sqrt{A} = \sqrt{A} \text{ whenever } A>0, \text{ with the } A=0 \text{ case trivial.}$$

$$B/\hat{\mu}^2 \leq B/(2B)^{2/3} = 2^{1/3} B^{1/3} \text{ whenever } B>0, \text{ with the } B=0 \text{ case trivial.}$$

Therefore

$$\hat{\mu} + A/\hat{\mu} + B/\hat{\mu}^2$$

$$\leq \sqrt{A} + (2B)^{1/3} + \sqrt{A} + 2^{1/3} B^{1/3}$$

$$= 2\sqrt{A} + [2^{1/3} + 2^{1/3}] B^{1/3}$$

$$= 2\sqrt{A} + (3/2) \cdot 2^{1/3} B^{1/3}.$$

Factor-two guarantee:

Dropping  $B/\mu^2$  gives

$$C_{\text{min}} - \beta \geq \inf_{\{\mu>0\}} (\mu + A/\mu) = 2\sqrt{A}.$$

Dropping  $A/\mu$  gives

$$C_{\text{min}} - \beta \geq \inf_{\{\mu>0\}} (\mu + B/\mu^2) = (3/2) \cdot 2^{1/3} B^{1/3}.$$

Hence  $C_{\text{min}} - \beta$  is at least the maximum of these two nonnegative quantities. Since  $\max\{x, y\} \geq (x+y)/2$ ,

$$C_{\text{closed}} - \beta = x + y \leq 2 \max\{x, y\} \leq 2(C_{\text{min}} - \beta).$$

The derivative of  $C$  is

$$C'(\mu) = 1 - A/\mu^2 - 2B/\mu^3.$$

The exact optimizer for  $A+B>0$  is the unique positive root of  $\mu^3 - A\mu - 2B = 0$ . The closed majorant avoids solving this cubic.

### 4. Evidence and review classification

Primary source: AO48-DER-011-v1.0 §1–§2, Drive ID 1688uChZmGCLjnOtg4ERpaS-kK0kGqzQM.

Underlying threshold source: GP-DER-037-v1.0, Drive ID 15pI0fJJXeMkbwSxaMipc8lod3PeQ9jZc\_Ju9mN6vUCQ.

Closure census: AO48-REG-018-v1.0 S6 identified one independent rederivation or numerical spot-check as the remaining step.

Second-line route completed here: OpenAI independently rederived the  $\hat{\mu}$  upper bound, the factor-two approximation guarantee, both exact endpoint cases, and the  $A=B=0$  bypass from the declared scalar objective.

Independence classification: strong dependent verification because the proposed formula was visible, but every inequality and minimization was reconstructed from the primitive objective.

## 5. Scope boundary and forbidden extrapolations

This proposal closes only the declared scalar optimization and majorization facts.

It does not by itself establish:

- • the invariant-strip lemma for a different flow or coefficient convention;
- • that the source jet-defined A and B are always finite or correctly bounded;
- • the joint central-transit criterion;
- • endpoint capture or sink capture;
- • finite-r remainder control;
- • adjacency, pair-Palm, or elder-rule pairing;
- • P0.2, Theorem B, any numerical theorem constant, or any machine-root transition.

The factor-two statement applies after subtracting the common additive base  $\beta$ . It must not be quoted as  $C_{\text{closed}} \leq 2C_{\text{min}}$  without checking the sign and role of  $\beta$ .

## 6. Dependencies affected

If approved, this closure:

- • retires repeated one-variable optimization and cubic-root calculations for this threshold;
- • permits use of a fully explicit sufficient threshold;
- • records exactness on both single-term endpoints  $A=0$  and  $B=0$ ;
- • records the degenerate  $A=B=0$  bypass;
- • permits downstream artifacts to cite the factor-two guarantee only for the declared scalar objective.

## 7. Dependencies explicitly unaffected

Unaffected:

- • validity of GP-DER-037's dynamical strip proof outside the scalar optimization step;
- • stored-escape adjudication and AO48's one-repair precommitment;
- • central transit, endpoint capture, and exact-field transfer;
- • P0.2, Theorem B, canonical theorem status, and active q0 machine state.

## 8. Coupled-advancement response

Mathematical status change proposed: the closed-form threshold and approximation guarantee move from a single-source derivation to second-line-verified narrow closure.

Architectural response required: retain explicit fields for the additive base, A/B domain, endpoint cases, degenerate bypass, and the distinction between strip-threshold optimization and the separate joint transit optimization.

No broader architecture change is required because current object-card, dependency, scope, and closure controls represent this transition.

Failure that would reveal insufficient controls:

- • a downstream artifact applies  $\mu=0$  when  $A=B=0$ ;
- • the factor-two guarantee is quoted without subtracting  $\beta$ ;
- • the threshold is used for a different scalar objective or for the joint transit optimization;
- • this closure is cited as proof of full capture or P0.2.

## 9. Human approval requirement

Human approval is REQUIRED because terminal mathematical closure remains reserved to Dylan M. Roy.

Suggested approval statement:

“I approve GP-CLS-PROP-015-v1.0 exactly as a closure of the declared scalar strip-threshold majorant, its factor-two guarantee after subtracting the common base, its exact  $A=0$  and  $B=0$  endpoint cases, and its  $A=B=0$  bypass. Record the terminal label CLOSED — EXACT CLOSED STRIP-THRESHOLD MAJORANT, preserve all provenance, and do not infer the joint transit criterion, endpoint capture, P0.2, Theorem B, or any machine-state result.”

## 10. Additive correction and reopening path

Reopening requires new contradictory evidence identifying an inequality error, minimization error, endpoint error, base-term misuse, or object mismatch. Corrections must preserve this proposal and all sources, map affected and unaffected dependencies, and reopen only this narrow scalar object unless broader contamination is demonstrated.

## Part III - Terminal-State Reconciliation

### 1. Closure decision

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Exact internal disposition</b>          | CLOSED - EXACT CLOSED STRIP-THRESHOLD MAJORANT                                                                    |
| <b>Closure date</b>                        | 2026-07-21                                                                                                        |
| <b>Evidence basis</b>                      | One-variable inequality, endpoint analysis, and degenerate-case bypass.                                           |
| <b>Independent review already recorded</b> | Independent random and endpoint tests confirmed the displayed bound and its restricted factor-two interpretation. |

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

### 2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

### 3. Limits

- No dynamical strip theorem or Palm/probability consequence without separate hypotheses.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

## Part IV - Reproduction Instructions

5. Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
6. Reconstruct the exact statement from Part I without relying on the historical verdict.
7. Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
8. Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
9. Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

## Part V - Reviewer Verdict Form

|                                      |                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------|
| <b>Reviewer</b>                      | _____                                                                           |
| <b>Affiliation / expertise</b>       | _____                                                                           |
| <b>Prior exposure</b>                | None / limited / substantial - explain: _____                                   |
| <b>Independent reconstruction</b>    | Complete / partial / not possible                                               |
| <b>Verdict</b>                       | APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY |
| <b>Exact scope accepted</b>          | _____                                                                           |
| <b>Critical or material findings</b> | _____<br>_____                                                                  |
| <b>Reproduction artifacts</b>        | Code/log/hash/location: _____                                                   |

**Signature and date**

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

## Part VI - Manifest and Provenance

**Controlling source:** <https://docs.google.com/document/d/1DHOrg2nEpe2gSSyQTFuijdoOvh80For06TZ7aQAhu-o/edit>

**Closure / terminal record:** [https://docs.google.com/document/d/1Zm65j9BZe\\_z3P\\_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit](https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit)

Local source snapshot used to construct this packet: GP-CLS-PROP-015-v1.0 — Proposed terminal closure of the closed strip-threshold majorant

Local source snapshot SHA-256 (DOCX export, transport checksum only):  
a72de3027baa875114eafc24681057eb6300f8392d2bce256457ed13e572ac06

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.